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Editorial Office
Journal of Statistical Software

Dear Editors,

We are pleased to submit our manuscript entitled “**StickForStats: A Statistical Analysis Platform with Automatic Assumption Validation**” for consideration for publication in the *Journal of Statistical Software*.

*Summary

StickForStats is an open-source statistical analysis platform that addresses a fundamental problem in research: statistical assumption violations often go unchecked, leading to unreliable results. Our platform introduces the Guardian system—the first implementation of *mandatory, automatic* assumption validation integrated directly into statistical software.

*Key Contributions

1. **Guardian System:** Automatic assumption validation with eight validators (normality, variance homogeneity, independence, linearity, outliers, sample size, multicollinearity, publication bias) that cannot be bypassed, ensuring researchers always see assumption status before results.
2. **Design Contract:** A novel software architecture principle where “no statistical result may exist without an explicit, traceable assumption context.” This is enforced through 93 automated tests (38 backend, 55 frontend).
3. **AI-Powered Advisor:** Natural language guidance for test selection and automatic generation of publication-ready methods sections following APA/JARS guidelines.
4. **Paper Parser:** Manuscript analysis tool that detects statistical reporting errors and assesses reproducibility.
5. **Comprehensive Validation:** All statistical functions validated against SciPy (14+ digit agreement) and cross-validated against R. Case studies use real datasets (Fisher’s Iris, UCI Wine Quality) with reproducible results.

*Relevance to JSS

This manuscript is highly relevant to JSS because:

- It presents novel statistical software with a unique approach to assumption checking
- The software is open-source and freely available at https://github.com/visvikbharti/stickforstats_new

- The paper includes comprehensive code examples and a complete replication package
- All numerical results are reproducible using provided scripts

*Replication Materials

The complete replication package is included with the submission:

- `run_all_validations.py` – SciPy validation (all tests pass)
- `validate_against_R.R` – R cross-validation (all tests pass)
- `verify_case_studies_FINAL.py` – Case study verification
- `data/winequality-red.csv` – Downloaded UCI Wine Quality dataset

*Declarations

- This manuscript has not been published elsewhere and is not under consideration by another journal.
- All authors have approved the manuscript and agree with its submission.
- There are no conflicts of interest to declare.
- The software is released under an open-source license.

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We believe StickForStats represents a significant contribution to statistical software and will be of great interest to the JSS readership. We look forward to your response.

Sincerely,

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